Vishal Khan

AI / Machine Learning Engineer

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LinkedIn | GitHub | AI Twin: live demo

Chat with my AI Twin

Right to work in the UK. Part-time now; full-time from January 2027 on the Graduate Route, which runs 18 months and needs no sponsor licence to hire me. After that, eligible to switch to Skilled Worker as a new entrant.

Four years in finance and data engineering before ML. I ship LLM systems to production: a live RAG assistant on **FastAPI**, **ChromaDB** and **Azure OpenAI** with intent-boosted retrieval, SSE streaming, tool calling, retrieved text fenced off against prompt injection and about **390 tests** in CI; and a multi-agent pricing system over SentenceTransformer embeddings. Previously built **Databricks** pipelines over millions of telecom records at Teleperformance, cutting query time and infrastructure cost by ∼**20%** each. MSc AI & Computer Science at Northeastern University London, completing January 2027; prior MSc thesis benchmarked DQN, A2C and PPO for dynamic pricing.

SELECTED ENGINEERING WORK

AI Digital Twin: Production RAG Assistant Live | Source

- **Objective:** Answer recruiter and interviewer questions from grounded, cited evidence, including live GitHub data a static knowledge base cannot hold.
- **Approach:** FastAPI, ChromaDB and Azure OpenAI with **8** LLM-callable tools and three answer modes; every answer lists its sources underneath, and interview mode also cites each claim inline. Source-aware retrieval applies per-source distance thresholds before preference boosting, so a preferred source wins near-ties but cannot pull an irrelevant chunk into the prompt; follow-ups fold the last two turns into the embedding query, holding subject continuity without a second LLM call. Auto-ingested GitHub READMEs are delimited as untrusted data against prompt injection. **React 19** and **TypeScript** front end streaming over SSE.
- **Result:** Live on **Azure Container Apps** after migrating off Railway, with a GitHub Actions to GHCR image pipeline at ∼**$4.70/month**. About **390 tests** across backend and frontend under `mypy` strict, `ruff` and `oxlint` in CI.

Multi-Agent Pricing System Source

- **Objective:** Estimate product prices from unstructured descriptions, and test whether a fine-tuned open model can match frontier LLMs on the task.
- **Approach:** Ensemble of specialist agents with a coordinator reconciling their outputs; RAG over SentenceTransformer embeddings in Chroma; frontier agent migrated from the OpenAI SDK to Azure OpenAI, resolving a stale-module authentication failure. Benchmarked **16 models** against a human baseline on one evaluation harness: classical ML through XGBoost, five frontier LLMs, and fine-tuned open models.
- **Result:** Best model, a fine-tuned 4-bit **Llama 3.2**, cut error **62%** below the constant baseline and beat every frontier model tested, including **GPT-5.1** by **11%**, taking the same base model from worse than the baseline to the best result tested through fine-tuning alone.

EXPERIENCE

Teleperformance Global Business Pvt. Ltd. [letter] *June 2024 to Jan 2026*
Data Engineer

- Built and maintained **Databricks** pipelines processing millions of telecom records (CDRs, billing, agent performance), serving a data engineering team of **50+** across multiple regions.
- Migrated on-prem SQL Server schemas to Databricks, improving query performance by ∼**20%** and cutting infrastructure cost by ∼**20%**.
- Owned **9 pipelines** end to end on Databricks Workflows, translating ambiguous stakeholder requirements into scheduled jobs that retired **30+** recurring manual reports.

Avant Garde Clean Room & Engg. Solutions Pvt. Ltd. [letter] *Aug 2021 to Dec 2023*
Executive, Commercial & Sales Analytics

- Built ML sales forecasting models achieving **10-20% MAPE** on monthly revenue projections, used daily for commercial planning.
- Turned ad-hoc commercial questions into recurring Python and SQL reporting workflows the team relied on.
- Rebuilt revenue and collections reporting in Excel (macros, VBA), cutting the monthly cycle from ∼**2 hours** to ∼**30 minutes**.

Earlier: finance, accounts and taxation at Aryan Care Pvt. Ltd. (2021) and PJRJ Associates (2018 to 2020). Statutory filings, audit support, reconciliation.

TECHNICAL SKILLS

Languages	Python, SQL, TypeScript, JavaScript, R, HTML/CSS
LLM & Agents	RAG, prompt engineering, agentic systems, MCP, LangChain, tool calling, Azure OpenAI, prompt-injection defence, evaluation harnesses
Machine Learning	Deep learning, reinforcement learning, classification, regression, time-series forecasting, NLP, feature engineering, model evaluation
Libraries	PyTorch, TensorFlow, Keras, scikit-learn, SentenceTransformers, Pandas, NumPy, Statsmodels, Matplotlib, Plotly
Engineering	FastAPI, Pydantic, ChromaDB, REST, SSE streaming, React 19, Vite, Tailwind, Docker, Azure Container Apps, GitHub Actions CI, GHCR, Vercel, Railway, Streamlit
Testing & QA	pytest, vitest, Testing Library, mypy (strict), ruff, oxlint
Data & Cloud	Databricks, Databricks Workflows, Azure, AWS, ETL pipelines, pipeline orchestration, SQL Server, MySQL, MongoDB, Neo4j
BI	Power BI, Tableau, Excel

EDUCATION

Northeastern University London, UK [cert] *Jan 2026 to Jan 2027*
MSc Artificial Intelligence & Computer Science | **95/100** in Programming for Data Applications

Liverpool John Moores University, UK [cert] *Mar 2023 to Mar 2024*
MSc Data Science

IIIT Bangalore, India [cert] *Jan 2022 to Jan 2023*
Executive PG Programme in Data Science

University of Delhi, India [cert] *Nov 2018 to Nov 2021*
BCom (Commerce)

RESEARCH

MSc Thesis: Reinforcement Learning for Dynamic Pricing in E-commerce [pdf] *LJMU, 2024*

- **Objective:** Test whether RL agents outperform rule-based pricing policies in a simulated e-commerce environment.
- **Approach:** Built a custom Gym-style environment on real e-commerce data, then trained and compared DQN, A2C and PPO agents across multi-episode simulations.
- **Result:** **DQN** produced the most stable profit curve; wrote up the comparative analysis as the dissertation.

Current dissertation (Northeastern London, due 2027): agentic AI, focused on agent reliability and evaluation tooling.